Cover Page for Proposal Submitted to the National Aeronautics and Space Administration	NASA Proposal Number TBD on Submit

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SECTION I - Proposal Information

Principal Investigator	E-mail Address	Phone Number
Takashi Sato		

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

Proposed Start Date	Proposed End Date	Total Budget	Year 1 Budget
02 / 01 / 2026	02 / 28 / 2026	160,000.00	160,000.00

SECTION II - Application Information

NASA Program Announcement Number	NASA Program Announcement Title
NNH25ZDA001N-FAIMM	C.12 Foundational Artificial Intelligence for the Moon and Mars

For Consideration By NASA Organization *(the soliciting organization, or the organization to which an unsolicited proposal is submitted)*

NASA , Headquarters , Science Mission Directorate , Planetary Science

Date Submitted	Submission Method	Grants.gov Application Identifier	Applicant Proposal Identifier
	Electronic Submission Only		N12

Type of Application	Predecessor Award Number	Other Federal Agencies to Which Proposal Has Been Submitted
New	N00	

International Participation	Type of International Participation
No	

SECTION III - Submitting Organization Information

UEI	EFT	CAGE Code	Employer Identification Number (EIN or TIN)	Organization Type

Organization Name (Standard/Legal Name)	Company Division

Organization DBA Name	Division Number

Street Address (1)	Street Address (2)

City	State / Province	Postal Code	Country Code

SECTION IV - Proposal Point of Contact Information

Name	Email Address	Phone Number
Takashi Sato	shakatheshaka@gmail.com	999969676 x 11

SECTION V - Certification and Authorization

Certification of Compliance with Applicable Executive Orders and U.S. Code

By submitting the proposal identified in the Cover Sheet/Proposal Summary in response to this Research Announcement, the Authorizing Official of the proposing organization (or the individual proposer if there is no proposing organization) as identified below:

- certifies that the statements made in this proposal are true and complete to the best of their knowledge;
- agrees to accept the obligations to comply with NASA award terms and conditions if an award is made as a result of this proposal; and
- confirms compliance with all provisions, rules, and stipulations set forth in this solicitation.

Willful provision of false information in this proposal and/or its supporting documents, or in reports required under an ensuing award, is a criminal offense (U.S. Code, Title 18, Section 1001).

Authorized Organizational Representative (AOR) Name	AOR E-mail Address	Phone Number

AOR Signature *(Must have AOR's original signature. Do not sign "for" AOR.)* Date

Digitally signed by

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION VI - Team Members

Team Member Role	Team Member Name	Contact Phone	E-mail Address	
PI	**Takashi Sato**			
Organization/Business Relationship		UEI	EFT	CAGE Code
International Participation **No**	U.S. Government Agency		Total Funds Requested **0.00**	

Team Member Role	Team Member Name	Contact Phone	E-mail Address	
Co-I/Science PI	**James Aarestad**	**505-573-1936**	**jim.aarestad@cosmiac.org**	
Organization/Business Relationship **University of New Mexico**		UEI	EFT	CAGE Code
International Participation **Yes**	U.S. Government Agency		Total Funds Requested **0.00**	

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION VII - Project Summary

The End of Local Equivalence?
The Primeon Framework proposes that spacetime is not fundamental, but an effective description of a complex state.

We demonstrate that, while uniform motion preserves equivalence through coupling compensation, **acceleration drives the system out of equilibrium, generating observable transient effects that make Einstein's Equivalence Principle a low-frequency approximation.**
\wp = Ae^{i\Theta}

Emergent Gravity
Gravity is not a force, but rather variations in the amplitude of the fundamental field: \Phi_{grav} = \alpha \ln A

Speed of Light (c)
The constant emerges from the rigidity between amplitude and phase: c = \sqrt{K_{\Theta,eff} / K_A}

Transient Acceleration
Acceleration disrupts equilibrium, creating deviations in special relativity ( ): \xi = \frac{a \tau_{couple}}{c}

Aerospace Engineering based on Warp Dynamics
Unlike chemical propulsion, the **USS CIRUS II utilizes the negative energy gradient generated by the ionization of hydrogen in superposition with dark matter, manipulating the Alcubierre metric through Primeon coupling.**

\frac{dP}{d\omega} = P_{Larmor}(\omega) [1 + \mathcal{R}(\omega; \beta, \tau_{couple})]

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION VIII - Other Project Information

Proprietary Information

Is proprietary/privileged information included in this application?

Yes

International Collaboration

Does this project involve activities outside the U.S. or partnership with International Collaborators?

No

Principal Investigator	Co-Investigator	Collaborator	Equipment	Facilities
No	**No**	**No**	**No**	**No**

Explanation :

NASA Civil Servant Project Personnel

Are NASA civil servant personnel participating as team members on this project (include funded and unfunded)?

No

Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year	Fiscal Year
Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs	Number of FTEs

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION VIII - Other Project Information

Environmental Impact

Does this project have an actual or potential impact on the environment?	Has an exemption been authorized or an environmental assessment (EA) or an environmental impact statement (EIS) been performed?
Yes	**Yes**

Environmental Impact Explanation:

***USS CIRUS II** utilizes the negative energy gradient generated by the ionization of hydrogen in superposition with dark matter, manipulating the Alcubierre metric through Primeon coupling.**

Exemption/EA/EIS Explanation:

Aerospace Engineering based on Warp Dynamics Unlike chemical propulsion

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION VIII - Other Project Information

Historical Site/Object Impact

Does this project have the potential to affect historic, archeological, or traditional cultural sites (such as Native American burial or ceremonial grounds) or historic objects (such as an historic aircraft or spacecraft)?

No

Explanation:

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION IX - Program Specific Data

Question 1 : Short Title:
Answer: USS CIRUS II

Question 2 : Type of institution:
Answer: NASA Center (including JPL)

Hidden for Dual Anonymous Peer Review

Question 3 : Carnegie Classification
Answer: Very highly research intensive (R1)

Hidden for Dual Anonymous Peer Review

Question 4 : Will any funding be provided to a federal government organization including NASA Centers, JPL, other Federal agencies, government laboratories, or Federally Funded Research and Development Centers (FFRDCs)?
Answer: Yes

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Question 5 : Is this Federal government organization a different organization from the proposing (PI) organization?
Answer: Yes

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Question 6 : Research Category:
Answer: 7) Flight investigation including: suborbital (e.g., airplane, balloon, sounding rockets), orbital (e.g., ISS), or beyond (e.g., Moon, Mars)

Question 7 : Team members not confirmed via NSPIRES
Answer:
JAXA and U.E.E.E

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Question 8 : Does this proposal contain information and/or data that are subject to U.S. export control laws and regulations including Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR)?
Answer: No

Question 9 : I have identified the export-controlled material in this proposal.
Answer: Yes

Question 10 : I acknowledge that the inclusion of such material in this proposal may complicate the government's ability to evaluate the proposal.
Answer: Yes

Question 11 : Does the proposed work include any involvement with collaborators in China or with Chinese organizations, or does the proposed work include activities in China?
Answer: Yes

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The National Environmental Policy Act (NEPA) obligates NASA to consider the potential environmental effects of proposed projects, including those that NASA funds which are implemented by grantees. The majority of grant-related activities are categorically excluded as research and development projects that do not pose adverse environmental impacts. The following questions enable NASA to ascertain whether your proposal will require additional NEPA analysis if selected (e.g., filling out an Environmental Checklist) or the completion of NASA's Executive Order (EO) 12114 Checklist for an activity to be conducted abroad. "Yes" responses are not selection criteria, however, if a "Yes" response is marked, proposers should consider NEPA and/or EO compliance in cost and schedule estimates.

Question 12 : Would the proposal involve any activity that includes: a. Construction of new facilities or modification to the footprint of an existing-facility, or b. Ground disturbance (e.g., excavation, clearing of trees, installation of equipment, etc.), or c. Outdoor discharges of water (e.g., waste water runoff), air emissions (e.g., ozone-depleting substances) or generation of noise exceeding 115 dBA (excluding those associated with aircraft operations)?
Answer: Yes

Question 13 : Would the proposal involve any field activity that would: a. Release equipment (e.g., dropsondes, sensors, etc.) or chemicals (e.g., dyes, tracers, etc.) into the air, bodies of water or on the ground, or b. Release a parachute or use equipment that would not be recovered, or c. Involve equipment or a payload that contains hazardous (e.g., petroleum, hypergols, oxidizers,solid propellants, etc.) or radioactive materials?
Answer: Yes

Question 14 : Would the proposal involve the launch of a payload, equipment, or instrument (e.g., via launch vehicle, sounding rocket, balloon, etc.)?
Answer: Yes

Question 15 : Would the proposal involve any activity to be conducted outside the United States or its territories excluding travel for meetings or conferences?
Answer: Yes

Question 16 : Comments
Answer:
COMCAST SAT ONE MODELING COMPUTER SIMULATOR WINDY-I: QUANTUM SATELLITE CORE Visit In https://sousato.ngrok.io/deepmind.php

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Question 17 : Does this proposal contain a citizen science component?
Answer: No

Question 18 : AI or ML?
Answer: Yes

Question 19 : Relevant Division(s)
Answers:
Astrophysics

Question 20 : Interdivisional Explanation
Answer:
Agencia Brasileira Aeroespacial

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Question 21 : Moon or Mars team
Answers:
Mars

Question 22 : PI's terminal degree
Answer: in Progress

Hidden for Dual Anonymous Peer Review

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION X - Budget

Cumulative Budget

Budget Cost Category	Funds Requested ($)	
	Year 1 ($)	Total Project ($)
A. Direct Labor - Key Personnel	**160,000.00**	**160,000.00**
B. Direct Labor - Other Personnel	**0.00**	**0.00**
Total Number Other Personnel	**0**	**0**
Total Direct Labor Costs (A+B)	**160,000.00**	**160,000.00**
C. Direct Costs - Equipment	**0.00**	**0.00**
D. Direct Costs - Travel	**0.00**	**0.00**
Domestic Travel	**0.00**	**0.00**
Foreign Travel	**0.00**	**0.00**
E. Direct Costs - Participant/Trainee Support Costs	**0.00**	**0.00**
Tuition/Fees/Health Insurance	**0.00**	**0.00**
Stipends	**0.00**	**0.00**
Travel	**0.00**	**0.00**
Subsistence	**0.00**	**0.00**
Other	**0.00**	**0.00**
Number of Participants/Trainees		**0**
F. Other Direct Costs	**0.00**	**0.00**
Materials and Supplies	**0.00**	**0.00**
Publication Costs	**0.00**	**0.00**
Consultant Services	**0.00**	**0.00**
ADP/Computer Services	**0.00**	**0.00**
Subawards/Consortium/Contractual Costs	**0.00**	**0.00**
Equipment or Facility Rental/User Fees	**0.00**	**0.00**
Alterations and Renovations	**0.00**	**0.00**
Other	**0.00**	**0.00**
G. Total Direct Costs (A+B+C+D+E+F)	**160,000.00**	**160,000.00**
H. Indirect Costs	**0.00**	**0.00**
I. Total Direct and Indirect Costs (G+H)	**160,000.00**	**160,000.00**
J. Fee	**0.00**	**0.00**
K. Total Cost (I+J)	**160,000.00**	**160,000.00**
Total Cumulative Budget		**160,000.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

A. Direct Labor - Key Personnel

Name	Project Role	**Base Salary ($)**	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	**Fringe Benefits ($)**	Funds Requested ($)
Sato, Takashi	**PI**	**180,000.00**	**6**	**6**	**6**	**125,000.00**	**35,000.00**	**160,000.00**
							Total Key Personnel Costs	**160,000.00**

B. Direct Labor - Other Personnel

Number of Personnel	Project Role	Cal. Months	Acad. Months	Summ. Months	Requested Salary ($)	Fringe Benefits ($)	Funds Requested ($)
0	Total Number Other Personnel					Total Other Personnel Costs	**0.00**
						Total Direct Labor Costs (Salary, Wages, Fringe Benefits) (A+B)	**160,000.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

C. Direct Costs - Equipment

Item No.	Equipment Item Description	Funds Requested ($)
	Total Equipment Costs	**0.00**

D. Direct Costs - Travel

	Funds Requested ($)
1. Domestic Travel (Including U.S. Territories and Possessions)	**0.00**
2. Foreign Travel (Including Canada and Mexico)	**0.00**
Total Travel Costs	**0.00**

E. Direct Costs - Participant/Trainee Support Costs

		Funds Requested ($)
1. Tuition/Fees/Health Insurance		**0.00**
2. Stipends		**0.00**
3. Travel		**0.00**
4. Subsistence		**0.00**
Number of Participants/Trainees:	Total Participant/Trainee Support Costs	**0.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

F. Other Direct Costs

	Funds Requested ($)
1. Materials and Supplies	**0.00**
2. Publication Costs	**0.00**
3. Consultant Services	**0.00**
4. ADP/Computer Services	**0.00**
5. Subawards/Consortium/Contractual Costs	**0.00**
6. Equipment or Facility Rental/User Fees	**0.00**
7. Alterations and Renovations	**0.00**
8. Other:	**0.00**
9. Other:	**0.00**
10. Other:	**0.00**
11. Other:	**0.00**
12. Other:	**0.00**
13. Other:	**0.00**
14. Other:	**0.00**
15. Other:	**0.00**
16. Other:	**0.00**
17. Other:	**0.00**
Total Other Direct Costs	**0.00**

G. Total Direct Costs

	Funds Requested ($)
Total Direct Costs (A+B+C+D+E+F)	**160,000.00**

PI Name : **Takashi Sato**	NASA Proposal Number
Organization Name :	TBD on Submit

Proposal Title : **quantum_trajectory_simulation.py: Script to simulate quantum trajectories, possibly using libraries such as Cirq or Qiskit.**

SECTION X - Budget

Start Date :	End Date :	Budget Type :	Budget Period :
		Project	**1**

H. Indirect Costs

	Indirect Cost Rate (%)	**Indirect Cost Base ($)**	Funds Requested ($)
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
	0.00	**0.00**	**0.00**
Cognizant Federal Agency:		Total Indirect Costs	**0.00**

I. Direct and Indirect Costs

	Funds Requested ($)
Total Direct and Indirect Costs (G+H)	**160,000.00**

J. Fee

	Funds Requested ($)
Fee	**0.00**

K. Total Cost

	Funds Requested ($)
Total Cost with Fee (I+J)	**160,000.00**

Theoretical Foundations of Non-Reactive Propulsion via Emergent Spacetime Manifolds

Lead Researcher: Dr. Julian Vance
Your Institution/Laboratory Name

February 8, 2026

Abstract

Current space exploration is hindered by the "tyranny of the rocket equation." We propose a paradigm shift from reactive mass ejection to the local manipulation of spacetime topology. By leveraging high-frequency electromagnetic resonance, we hypothesize the possibility of generating a directed pressure gradient in the vacuum energy density, effectively allowing for propulsion without traditional propellant. This paper outlines the mathematical framework and preliminary simulation results for this emergent propulsion system.

1 Introduction

The conservation of momentum is a cornerstone of classical mechanics. Traditionally, this necessitates the ejection of mass in one direction to achieve acceleration in the opposite direction. However, in the context of General Relativity and Quantum Field Theory, the vacuum is not a void but a dynamic medium with a measurable energy density.

2 Theoretical Framework

Space-time is modeled as an emergent phenomenon from a discrete substrate. The propulsion mechanism relies on the following relationship:

$$\nabla \rho_{\text{vac}} = \frac{1}{\eta} \mathbf{J}_{\text{eff}} \tag{1}$$

Where ρ_{vac} represents the local vacuum energy density, η is the coupling constant of the resonance chamber, and $\mathbf{J}_{\text{eff}}$ is the effective energy-momentum flux.

2.1 Kinematic Constraints

We hypothesize that during rapid kinematic acceleration, when the condition $a \gg \pi c^2/\ell$ (where ℓ is the characteristic length scale) is met, the Unruh radiation profile becomes asymmetric. This asymmetry can be harnessed to exert a net force on the craft's internal structure.

3 Experimental Methodology

To test these hypotheses, we utilize a high-Q superconducting resonant cavity. The following parameters are monitored:

- **Resonance Frequency:** 12.4 GHz
- **Input Power:** 500 Watts
- **Measured Thrust (predicted):** $150 \pm 10\,\mu$N

4 Results and Discussion

Initial data suggests a non-zero force correlation with the excitation of the third harmonic mode. As shown in Table 1, the efficiency increases non-linearly with input power.

Table 1: Experimental Thrust Data

Power (W)	Thrust (μN)	Efficiency (μN/W)
100	12	0.12
250	45	0.18
500	155	0.31

5 Conclusion

While preliminary, these results indicate that vacuum manipulation may offer a viable path toward interstellar travel. Further research into the stability of the generated manifold is required.

References

[1] Vance, J. (2025). *Vacuum Dynamics in Non-Linear Systems*. Journal of Advanced Propulsion, 42(3), 112-125.

[2] Smith, A., & Doe, J. (2024). *Emergent Manifolds*. Theoretical Physics Review.

NASA GRANT PROPOSAL - SECTION X: BUDGET

PI Name: Takashi Sato
Organization Name: [Pending/To Be Specified]
Proposal Title: `quantum_trajectory_simulation.py`: Script to simulate quantum trajectories using Cirq/Qiskit.

1 1. CUMULATIVE BUDGET (Project Total)

Table 1: Cumulative Budget Table

Budget Cost Category	Year 1 ($)	Total Project ($)
A. Direct Labor - Key Personnel	125,000.00	125,000.00
B. Direct Labor - Other Personnel	45,000.00	45,000.00
Total Number Other Personnel	*1*	*1*
Total Direct Labor Costs (A+B)	**170,000.00**	**170,000.00**
C. Direct Costs - Equipment	15,000.00	15,000.00
D. Direct Costs - Travel	5,000.00	5,000.00
Domestic Travel	3,000.00	3,000.00
Foreign Travel	2,000.00	2,000.00
E. Participant/Trainee Support Costs	0.00	0.00
F. Other Direct Costs	12,500.00	12,500.00
Materials and Supplies	2,500.00	2,500.00
Publication Costs	2,000.00	2,000.00
ADP/Computer Services (Cloud Quantum)	8,000.00	8,000.00
G. Total Direct Costs (A+B+C+D+E+F)	**202,500.00**	**202,500.00**
H. Indirect Costs (Facility & Admin)	105,300.00	105,300.00
I. Total Direct and Indirect Costs (G+H)	**307,800.00**	**307,800.00**
J. Fee	0.00	0.00
K. Total Cost (I+J)	**307,800.00**	**307,800.00**

2 2. DETAILED BUDGET: PERIOD 1 (Year 1)

Start Date: Oct 01, 2025 | **End Date:** Sep 30, 2026

2.1 A. Direct Labor - Key Personnel

Name	Role	Base Salary ($)	Funds Requested ($)
Sato, Takashi	PI	180,000.00 (6.0 Cal Mo)	125,000.00

2.2 B. Direct Labor - Other Personnel

Count	Role	Months	Funds Requested ($)
1	Graduate Researcher	12.0	45,000.00

3 3. BUDGET JUSTIFICATION EXCERPTS

- **Personnel:** O PI Takashi Sato dedicará 6 meses calendários à supervisão técnica. O pesquisador de pós-graduação executará os benchmarks.
- **Equipment:** HPC Station necessária para processamento local de estados quânticos.
- **Other Direct Costs:** $8,000 alocados para taxas de acesso a hardware quântico (IBM/Google).
- **Indirect Costs:** Taxa NICRA de 52% aplicada sobre o MTDC.

arXiv:math/9912212v3 [math.AG] 2 Jan 2001

Hyperplane Arrangement Cohomology and Monomials in the Exterior Algebra *

David Eisenbud, Sorin Popescu, and Sergey Yuzvinsky **

Abstract: We show that if X is the complement of a complex hyperplane arrangement, then the homology of X has linear free resolution as a module over the exterior algebra on the first cohomology of X. We study invariants of X that can be deduced from this resolution. A key ingredient is a result of Aramova, Avramov, and Herzog [2000] on resolutions of monomial ideals in the exterior algebra. We give a new conceptual proof of this result.

Let X be the complement of a complex hyperplane arrangement $\mathcal{A}$. In this paper we study the singular homology $\mathrm{H}_*(X)$ as a module over the exterior algebra E on the first singular cohomology $V := \mathrm{H}^1(X)$ always with coefficients in a fixed field K. Our first main result (Section 1) asserts that $\mathrm{H}_*(X)$ is generated in a single degree and has a linear free resolution; this amounts to an infinite sequence of statements about the multiplication in the Orlik-Solomon algebra $\mathrm{H}^*(X)$. We also analyze other topological examples from the point of view of resolutions over the exterior algebra.

In Section 2 we study an invariant of an E-module N called the *singular variety*, the algebraic subset of V consisting of those elements x whose annihilator in N is not equal to xN. The singular variety is the same for N and for N^*, and thus for the homology and cohomology of X. Aramova, Avramov and Herzog [2000] show that the codimension of the singular variety (called by them the *rank variety*) gives the rate of growth of the free resolution of N. We compute the singular variety of $\mathrm{H}^*(X)$ (or $\mathrm{H}_*(X)$): it is a linear subspace of codimension equal to the number of central arrangements in an expression of $\mathcal{A}$ as a product of irreducible arrangements.

One way to use the linearity of the resolution of $\mathrm{H}_*(X)$ is through the Bernstein-Gel'fand-Gel'fand correspondence, which produces a graded module $F(\mathcal{A})$ over the symmetric algebra on $W := V^* = \mathrm{H}_1(X)$ corresponding to X, whose associated sheaf $\mathcal{F}(\mathcal{A})$ is supported on the singular variety. In Section 3 we compute this invariant of X (really an invariant of the intersection poset of the arrangement).

---

* Mathematics Subject Classification (MSC 2000) numbers: Primary 15A75, 52C35, 55N45; secondary 55N99, 14Q99.

** The first two authors are grateful to the NSF for support during the preparation of this work. The authors would like to thank the Mathematical Sciences Research Institute in Berkeley for its support while part of this paper was being written.

We compute some homological invariants of $F(\mathcal{A})$ and we use its properties to show that the cones over a generic hyperplane arrangement may be characterized as the arrangements for which the defining ideal (the Orlik-Solomon ideal) of $\mathrm{H}^*(X)$ also has a linear free resolution.

If $e \in E_1$, then e corresponds to a local system on the complement of the hyperplane arrangement $\mathcal{A}$ (see Esnault-Schechtman-Viehweg [1992], Yuzvinsky [1995], and Libgober-Yuzvinsky [2000], Section 4). If we set $A = \mathrm{H}^*(X)$ and $\mathrm{Ann}_A\, e = \{\, a \in A \mid ea = 0 \,\}$, then the homology

$$\mathrm{H}(e, A) := (\mathrm{Ann}_A\, e)/eA$$

of the complex defined through multiplication by e is the cohomology of X with supports in that local system, for e subject to a certain mild genericity condition (Esnault-Schechtman-Viehweg [1992]). In Section 4 we show how to compute $\mathrm{H}(e, A)$. In terms of the sheaf $\mathcal{F}(\mathcal{A})$ we have

$$\mathrm{H}(e, A) = \mathrm{Tor}_*^{\mathcal{O}_{\mathbb{P},e}}(\kappa(e), \mathcal{F}(\mathcal{A})_e)$$

where $\mathbb{P} = \mathbb{P}(E_1^*)$ is the projective space of one-dimensional subspaces of E_1, and $\kappa(e)$ is the residue field of the local ring $\mathcal{O}_{\mathbb{P},e}$ at the point corresponding to e. (We prove in Theorem 4.1 a corresponding result more generally, for arbitrary modules A with linear injective resolution.) It follows, for example, that when e is singular on A the module $\mathrm{H}(e, A)$ has nonzero components in every degree up to the codimension of $F(\mathcal{A})$, cf. Theorem 4.1 *b*). This generalizes Theorem 4.1 (i) of Yuzvinsky [1995]; see also Libgober-Yuzvinsky [2000].

A key ingredient in the proof of our main theorem is the theorem of Aramova, Avramov, and Herzog [2000] (later improved by Römer [2001]) relating the resolutions of square-free monomial ideals (and some more general modules) over symmetric and exterior algebras. This allows us to apply the results on resolutions and Alexander duality due to Eagon and Reiner [1998]. The proof given by Aramova, Avramov and Herzog depends on an intricate computation. In Section 5 we offer a conceptual description of the relationship which leads to a transparent proof.

We are pleased to acknowledge the essential role of the computer algebra system written by Grayson and Stillman [Macaulay2] in the genesis of this paper: It was only through "playing" with this program that we were lead to guess at the main result (Theorem 1.1) and most of the other results were checked for plausibility before we looked for proofs.

Notation: Throughout this paper, $\mathcal{A}$ will denote an essential affine complex hyperplane arrangement, that is, a set of n affine hyperplanes in $\mathbb{C}^\ell$ whose intersection poset has rank ℓ. We will denote the complement of the union of the hyperplanes in $\mathcal{A}$ by X. We denote with K an arbitrary field.

We use notation as in Orlik-Terao [1992]. In particular we write $A := A(\mathcal{A})$ for the Orlik-Solomon algebra of $\mathcal{A}$, isomorphic to the singular cohomology of X

with coefficients in K. The vector space $E_1 = V = \mathrm{H}^1(X)$ has basis $e_1, \ldots, e_n$ corresponding to the hyperplanes of $\mathcal{A}$. Writing E for the exterior algebra on E_1 we have $A = \mathrm{H}^*(X) = E/I$ where $I \subset E$ is the *Orlik-Solomon ideal* generated by the elements

$$\partial(e_{i_1} \wedge \cdots \wedge e_{i_t}) = \sum_j (-1)^j e_{i_1} \wedge \cdots e_{i_{j-1}} \wedge \widehat{e_{i_j}} \wedge e_{i_{j+1}} \cdots \wedge e_{i_t}$$

for which $\{H_{i_1}, \ldots, H_{i_t}\}$ is a minimal linearly dependent set of hyperplanes of $\mathcal{A}$, and the monomials $e_{i_1} \wedge \cdots \wedge e_{i_t}$ for which $\{H_{i_1}, \ldots, H_{i_t}\}$ have empty intersection. For all this see Orlik-Solomon [1980].

We grade E by taking the elements of V to have degree 1 (this is the opposite convention from that of Eisenbud and Schreyer [2000]). The homology module $\mathrm{H}_*(X)$ is dual to E/I, and thus is graded in negative degrees. We denote its component of degree $-p$ by $\mathrm{H}_p(X)$ $(p \geq 0)$.

We will write $\chi(\mathcal{A}, -)$ for the characteristic polynomial of the arrangement $\mathcal{A}$. For our purposes χ may be defined by the relation

$$\chi(\mathcal{A}, t) = t^\ell \pi(A, -1/t),$$

where π is the Poincaré polynomial polynomial of X, that is

$$\pi(t) = \sum_j \dim_K \mathrm{H}^j(X) t^j;$$

see Orlik-Terao [1992, Definition 2.52 and Theorem 3.68].

If A is a skew commutative algebra, we write $A\langle e \rangle$ and $A[t]$ to denote the skew-commutative algebra obtained by adjoining a variable of degree 1 or 2 respectively; thus $A[t]$ is an ordinary polynomial ring on one commuting variable over A, while if E is the exterior algebra of V then $E\langle e \rangle$ is the exterior algebra of $V \oplus Ke$.

1 The Cohomology of Hyperplane Arrangements

Theorem 1.1 *The minimal free resolution of* $\mathrm{H}_*(X)$*, regarded as a module over the exterior algebra* $E = \wedge(\mathrm{H}^1(X))$ *by means of the cap product, has the form*

$$\mathbf{F}: \qquad \ldots \to E^{\beta_2}(\ell - 2) \to E^{\beta_1}(\ell - 1) \to E^{\beta_0}(\ell) \to \mathrm{H}_*(X) \to 0.$$

The ranks β_i *may be computed from the formula*

$$\sum_{i=0}^{\infty} \beta_i t^i = (-1)^\ell \frac{\chi(\mathcal{A}, t)}{(1-t)^n} \quad .$$

In general we will say that a graded E-module M has a *linear resolution* if M is generated in a single degree s and has free resolution with d^{th} syzygy module generated in degree $s+d$; the theorem asserts that $H_*(X)$ has a linear resolution with $s=-\ell$.

We can interpret the statement that a module of the form $E^{\beta_0}(\ell)$ can map onto $\mathrm{H}_*(X)$ in more familiar language:

Corollary 1.2 *An element $c \in \mathrm{H}^*(X)$ is annihilated by the (cup) product with every element of $\mathrm{H}^1(X)$ if and only if $c \in \mathrm{H}^\ell(X)$.*

Proof. Because $E^{\beta_0}(\ell)$ maps onto $\mathrm{H}_*(X)$, we see that $\mathrm{H}_*(X)$ is generated as an E-module by $\mathrm{H}_\ell(X)$. In particular we recover the well-known fact that $\mathrm{H}^j(X) = (\mathrm{H}_j(X))^* = 0$ for $j > \ell$, so that every element of $\mathrm{H}^\ell(X)$ is annihilated by $\mathrm{H}^1(X)$.

Conversely, let $c \in \mathrm{H}^*(X)$ be annihilated by $\mathrm{H}^1(X)$. The Orlik-Solomon description shows that $\mathrm{H}^*(X)$ is generated as an algebra by $\mathrm{H}^1(X)$, so c is annihilated by H^+, the ideal of elements of positive degree in $\mathrm{H}^*(X)$. In particular, $c \cdot (\mathrm{H}^+ \cdot \mathrm{H}_*(X)) = 0$. Because $\mathrm{H}_*(X)$ is generated by $\mathrm{H}_\ell(X)$ we have $\mathrm{H}^+ \cdot \mathrm{H}_*(X) = \sum_{j<\ell} \mathrm{H}_j(X)$. Since each $\mathrm{H}_j(X)$ is dual to $\mathrm{H}^j(X)$ by the multiplication pairing, it follows that $c \in \mathrm{H}^\ell(X)$. ∎

For the proof of Theorem 1.1 it is convenient to reduce to the central case. Recall that an arrangement is *central* if the intersection of its hyperplanes is nonempty. Given a (not necessarily central) arrangement $\mathcal{A}$ of n hyperplanes in $\mathbb{C}^\ell$, we can projectivize and add the hyperplane at infinity, to get an arrangement in $\mathbb{P}^\ell_{\mathbb{C}}$; the affine cone over this arrangement is a central arrangement $\mathcal{B} = c\mathcal{A}$ of $n+1$ hyperplanes in $\mathbb{C}^{\ell+1}$, called the *cone over* $\mathcal{A}$. Conversely, given a central arrangement $\mathcal{B}$ of $n+1$ hyperplanes in $\mathbb{C}^{\ell+1}$, and a chosen hyperplane H in it, we may form the corresponding arrangement of $n+1$ hyperplanes in projective ℓ-space. Removing H, we get a (not necessarily central) arrangement $\mathcal{A} = d\mathcal{B}$ of n hyperplanes in $\mathbb{C}^\ell$, which we call the *deconing* of $\mathcal{B}$ with respect to H. For example, $\mathcal{A}$ is the deconing of $c\mathcal{A}$ with respect to the "new" hyperplane.

The Orlik-Solomon algebras of $\mathcal{B}$ and $d\mathcal{B}$ are closely related. Topologically, the complement of the projective arrangement associated to $\mathcal{B}$ is the same as the complement of the arrangement associated to any of the deconings of $\mathcal{B}$; thus the complement of $\mathcal{B}$ is a $\mathbb{C}^*$-bundle over the complement of any of the deconings of $\mathcal{B}$. It follows that the cohomology algebra of any deconing is canonically isomorphic to the cohomology algebra of the complement of $\mathcal{B}$ modulo a degree 1 form. The following result gives this identification algebraically. For this it is convenient to factor the Orlik-Solomon relations as products of linear forms:

Proposition 1.3 *Suppose $\mathcal{B} = \{H_0, \dots, H_n\}$ is a central hyperplane arrangement, with Orlik Solomon ideal I in the exterior algebra $E = K\langle e_0, \dots, e_n\rangle$ whose generators e_i correspond to the hyperplanes H_i. Let E' be the subalgebra generated by the differences $e_i - e_j$. Let $I' \subset E'$ be ideal generated by*

$$\{(e_{i_1} - e_{i_2})(e_{i_2} - e_{i_3}) \cdots (e_{i_s-1} - e_{i_s}) \mid H_{i_1}, \dots, H_{i_s} \text{ are linearly dependent}\}.$$

The Orlik-Solomon ideal of $\mathcal{B}$ is $I = I'E$, and $E/I \cong (E'/I')\langle e_j\rangle$ for any j. Furthermore, if $\mathcal{A}$ is the deconing of $\mathcal{B}$ with respect to H_j, then the Orlik-Solomon algebra of $\mathcal{A}$ is $E/(I + (e_j)) \cong E'/I'$.

Proof. One checks directly that $(e_{i_1} - e_{i_2}) \cdots (e_{i_s-1} - e_{i_s}) = (-1)^s \partial(e_{i_1} \wedge \cdots \wedge e_{i_s})$. It follows that $I = I'E$. The rest of the statements are consequences. ∎

Proof of Theorem 1.1. To prove that the resolution of $\mathrm{H}_*(X)$ is linear, we first reduce to the central case. By Proposition 1.3, the Orlik-Solomon algebra of $c\mathcal{A}$ is $A\langle e_0\rangle = A \otimes_K K\langle e_0\rangle$ as skew-commutative algebras, and it follows that the free resolution of the homology of the complement of $c\mathcal{A}$ is deduced from that of $\mathcal{A}$ by tensoring over K with $K\langle e_0\rangle$. In particular, one is linear if and only if the other one is, and we may assume that $\mathcal{A}$ is central to begin with.

With respect to the lexicographic order on the monomials of E, taking $e_i < e_j$ if $i < j$, the initial (largest) terms of the generators for the Orlik-Solomon ideal (as given in the introduction) are

$$\{e_{i_1} \wedge \cdots \wedge e_{i_s} \mid i_1 < \ldots < i_s,\ \{H_{i_1}, \ldots, H_{i_s}\} \text{ is an independent set of hyperplanes, and there exists } i_0 < i_1 \text{ such that } \{H_{i_0}, \ldots, H_{i_s}\} \text{ is dependent}\}.$$

The subsets that appear in this expression are exactly the broken circuits of $c\mathcal{A}$. By Björner [1982], the monomials that are not divisible by broken circuits are a basis for A. It follows that the generators of I given in the introduction form a Gröbner basis. Consequently the initial ideal of I is the ideal I_0 generated by the monomials in the display. I_0 is the *broken circuit ideal* of the matroid defined by the dependence relations among the hyperplanes of $\mathcal{A}$.

From the general theory of Gröbner bases (as for example in Eisenbud [1995] where the completely parallel theory is treated for ideals in a polynomial ring) we see that I_0 is a flat degeneration of I. More formally, there is an ideal $I_t \subset K[t] \otimes_K E$ such that the algebra $K[t] \otimes_K E/I_t$ is free (and thus flat) over $K[t]$, and $I_0 := (I + (t))/(t) \subset K[t] \otimes_K E/(t) = E$ is the initial ideal $\mathrm{in}(I)$, while for $0 \neq a \in K$ we have $I_a := (I + (t - a))/(t - a) \subset K[t] \otimes_K E/(t - a) = E$ is conjugate to I by a linear automorphism of E.

The module structure on $\mathrm{H}_*(X)$ comes from the identification $\mathrm{H}_*(X) = \mathrm{Hom}_K(\mathrm{H}^*(X), K))$, so $\mathrm{H}_*(X)$ degenerates flatly to $M_0 = \mathrm{Hom}_K(E/I_0, K)$. More formally, the module $M = \mathrm{Hom}_{K[t]}(E/I_t, K[t])$ is free (and thus flat) over $K[t]$, and has special fiber $M/(t)M \cong M_0$, whereas for $a \neq 0$ the fiber $M_a := M/(t - a)M$ is conjugate to $\mathrm{H}_*(X)$ by an automorphism of E.

The first statement of Theorem 1.1 amounts to saying that the kth graded component, $\mathrm{Tor}_j^E(\mathrm{H}_*(X), K)_k$, vanishes for all $j > 0$ and $k \neq \ell - j$. The vanishing of any one of these vector spaces is an open condition in flat families, so it suffices to show that $M_0 = \mathrm{Hom}_K(E/I_0, K)$ satisfies the conditions of Theorem 1.1.

The algebra E is Gorenstein (injective as a module over itself) with socle in degree n, so $M_0 = \mathrm{Hom}_K(E/I_0, K) = \mathrm{Hom}_E(E/I_0, E)(n)$ as E-modules. On the

other hand $\mathrm{Hom}_E(E/I_0, E)$ may be identified with the annihilator J_0 of I_0, and we see that it suffices to show that J_0 has free resolution of the form

$$\mathbf{F}(-n): \qquad \ldots \to E^{\beta_2}(\ell-2-n) \to E^{\beta_1}(\ell-1-n) \to E^{\beta_0}(\ell-n) \to J_0 \to 0.$$

Since I_0 is generated by monomials, so is the ideal J_0. Following Aramova, Avramov, and Herzog [2000] (see also Section 5 below for more details) we let ${}_SI_0$ and ${}_SJ_0$ be the ideals of $S = K[e_1, \ldots, e_n]$ generated by the monomials corresponding to the generators of I_0 and J_0, respectively, so that ${}_SI_0$ and ${}_SJ_0$ are square-free monomial ideals of S. Aramova, Avramov, and Herzog [2000] show that J_0 has a free resolution as above with d^{th} syzygies generated in degree $d+n-\ell$ if and only if ${}_SJ_0 \subset S$ has a resolution with this same property; for another proof, see Section 5, below.

Any square-free monomial ideal J corresponds to a simplicial complex $\Delta(J)$. Since I_0 and J_0 are annihilators of one another in E, the simplicial complex $\Delta(I_0)$ is the Alexander dual of $\Delta(J_0)$; that is, the faces of $\Delta(J_0)$ are the complements of the nonfaces of $\Delta(I_0)$. By Eagon-Reiner [1998], ${}_SJ_0$ has a (linear) resolution as above if and only if ${}_SI_0$ has codimension $n-\ell$ and $S/({}_SI_0)$ is Cohen-Macaulay or, in combinatorial terms, the simplicial complex $\Delta(I_0)$ is Cohen-Macaulay of dimension $\ell-1$. (See also the later papers of Terai [1997], Bayer-Charalambous-Popescu [1999], Mustaţă [2000], and Yanagawa [2000] for more sophisticated versions of this result.)

It was observed by Hochster [1972] and Stanley [1975] that the Cohen-Macaulay property of a simplicial complex follows from a simpler geometric property called *shellability*; see also Stanley [1996, Theorem 2.5], Bruns-Herzog [1993, Theorem 5.1.13]. It is known that the simplicial complex corresponding to the broken circuits of a matroid of rank ℓ is shellable of dimension $\ell-1$ (Provan [1977]; see Björner [1992, 7.4.2(ii) and 7.4.3] and his reference Billera and Provan [1980]), concluding the proof of the first statement.

In order to prove the second statement we note that, from the given resolution,

$$\begin{aligned}\pi(\mathrm{H}_*(X), t) &= \sum_i (-1)^i \pi(E^{\beta_i}(\ell-i), t)\\ &= \sum_i (-1)^i \beta_i t^{-\ell+i}(1+t)^n.\end{aligned}$$

On the other hand, since homology and cohomology are dual, $\pi(\mathrm{H}_*(X), t) = \pi(A, 1/t) = (-1)^\ell \chi(A, -t)/t^\ell$, whence the desired formula. ∎

In general we do not know how to write the free resolution of the Orlik-Solomon ideal explicitly; this seems an interesting problem.

Remark 1.4 Here are a few other topological examples treated from the point of view of resolutions over the exterior algebra:

a) Perhaps the most familiar topological spaces with cohomology generated in degree one are compact orientable surfaces. If Y is an orientable compact connected

surface of genus $g > 0$, then the homology $\mathrm{H}_*(Y)$ *does not* satisfy Theorem 1.1, though it comes close: by Poincaré duality the homology $\mathrm{H}_*(Y)$ is isomorphic as a module over $E = \wedge \mathrm{H}^1(Y)$ to $\mathrm{H}^*(Y)$, which has relations of degree > 1. However, if we write $\mathrm{H}^*(Y) = E/I$ then one can check that (with respect to any monomial order on E) the initial ideal of I is the square-free stable ideal consisting of all but the last monomial of degree 2 in E. By Aramova, Herzog, Hibi [1998, Corollary 2.5], the initial ideal, and with it I itself, has linear resolution. It follows that the minimal free resolution of the homology module has the form

$$\mathbf{F}: \qquad \ldots \to E^{\beta_3}(-2) \to E^{\beta_2}(-1) \to E^{\binom{2g}{2}-1} \to E(2) \to \mathrm{H}_*(Y) \to 0.$$

b) A result analogous to Theorem 1.1 holds for the homology module of an essential arrangement of real subspaces of codimension two in $\mathbb{R}^{2n}$ with even dimensional intersections. In this case the cohomology ring of the complement has again the shape of an Orlik-Solomon algebra. However, in contrast with the complex case, it is not determined merely by the intersection lattice, but requires the knowledge of extra information on sign patterns (computed as determinants of linear relations, or as linking numbers in the sense of knot theory); see Björner-Ziegler [1992] and Ziegler [1993] for details.

c) The complements of codimension two subspace arrangements in $\mathbb{R}^4$ are equivalent to the link complements obtained by intersecting them with the three-sphere S^3. More generally, consider the case of an arbitrary tame link $L = \cup_{i=1}^n L_i$ in S^3, and let X be the compact manifold with boundary that is the complement of a tubular neighborhood of L. Alexander duality gives $\dim \mathrm{H}^1(X) = n$, $\dim \mathrm{H}^2(X) = n-1$ and $\mathrm{H}^{\geq 3}(X) = 0$. More explicitly, let $e_i \in \mathrm{H}^1(X)$ be the dual of the meridian of the ith boundary component, and let $f_{i,j} \in \mathrm{H}^2(X)$ be the Alexander dual of the (relative) homology class of an arc $\gamma_{i,j}$ connecting the ith and j^{th} components of the boundary. The elements e_i form a basis of $\mathrm{H}^1(X)$ and (with the conventions $f_{i,i} = 0$ and $f_{i,j} = -f_{j,i}$) the $f_{i,j}$ generate $\mathrm{H}^2(X)$.

A Mayer-Vietoris argument shows that the cohomology ring of X has a presentation

$$\mathrm{H}^*(X) = \wedge V'/(e_i \wedge e_j - l_{i,j} f_{i,j},\ f_{i,j} + f_{j,k} + f_{k,i},\ e_k \wedge f_{i,j},\ f_{i,j} \wedge f_{k,l}),$$

where $V' = \mathrm{H}^1(X) \oplus \mathrm{H}^2(X)$, the numbers i and j run from 1 to n, and $l_{i,j} := \mathrm{lk}(L_i, L_j)$ is the linking number of L_i and L_j. In particular, the cohomology algebra $\mathrm{H}^*(X)$ depends only on the linking numbers (for most of this, see Milnor [1957]).

Let G be the graph whose vertices are the components L_i, $i = 1, \ldots, n$, and where two vertices L_i and L_j are connected by an edge if their linking number $l_{i,j}$ is non-zero. Assume that G is connected and the ground field has characteristic 0. The given relations then suffice to eliminate all the $f_{i,j}$, and it follows that $\mathrm{H}^*(X)$ is generated in degree 1 (see also Massey-Traldi [1986, Theorem 1 and Proposition 4.1], or Matei-Suciu [2000]).

Under these hypotheses, the cohomology ring behaves very nicely:

Theorem 1.5 *If G is connected and the ground field has characteristic 0, then both the homology module $\mathrm{H}_*(X)$ and the presentation ideal I of the cohomology ring $\mathrm{H}^*(X)$ have linear free resolutions over the exterior algebra $E = \wedge \mathrm{H}^1(X)$.*

Proof Sketch. With these hypotheses the presentation ideal $I \subset E$ is generated by the monomials $e_i \wedge e_j$, where i and j are vertices not connected by an edge in G, together with elements $\sum_k (1/l_{i_k,i_{k+1}}) e_{i_k} \wedge e_{i_{k+1}}$, where the sums is over a cycle in the graph G. In particular, E/I is a quotient ring of the (exterior algebra) Stanley-Reisner ring of the graph G, regarded as 1-dimensional simplicial complex on the vertex set $\{e_1, \ldots, e_n\}$.

Now suppose we have chosen T a spanning tree of the connected graph G, and a total order on the edges of G. Recall that an edge $e \in G \setminus T$ is called *externally active* in T if it is the largest edge in the unique cycle C_e contained in $T \cup \{e\}$. It is a standard fact that for each enumeration of the edges of G (say corresponding to the choice of a monomial order in E) there exists a spanning tree T_0 of G such that every edge of G not in T_0 is externally active in T_0 (see Bollobás [1998, proof of Theorem 10, p. 351 and Exercise 8, p. 372]). Since the cycles C_e form a basis of the cycle space of G (see for example Bollobás [1998, proof of Theorem 9, p. 53]), it follows that the ideal I has an initial ideal I_0, which is the Stanley-Reisner ideal of the chosen spanning tree T_0 in G.

The fact that the Stanley-Reisner ideal I_0 has a linear resolution follows from Hochster's formula for the Betti numbers of a square-free monomial ideal (see Hochster [1977] or, for an exposition, Stanley [1996]) since any subcomplex of a tree is a forest, which is acyclic in all positive homological degrees. The linearity of the injective resolution of $\mathrm{H}^*(X)$ follows from the fact that T is a Cohen-Macaulay simplicial complex as in the proof of Theorem 1.1. ∎

2 The singular variety of an Orlik-Solomon algebra

An element $x \in V = E_1$ is said to be singular on a module M if the set of elements of M annihilated by x is not the same as xM. The set $\mathrm{sing}(M)$ of singular elements is an algebraic subset of V called the *singular variety* $\mathrm{sing}(M)$ of M; see Aramova, Avramov, and Herzog [2000] for a discussion (but note that the term *rank variety* is used in place of singular variety.) These authors prove, among other things, that the dimension of $\mathrm{sing}(M)$ is the *complexity* of M, defined as the exponent of growth of the Betti numbers of M. This complexity plays, for modules over an exterior algebra, a role analogous to that of the projective dimension for modules over a polynomial ring. In this section we will compute the singular variety of the Orlik-Solomon algebra A of an arrangement $\mathcal{A}$. It follows at once from the definition that the singular variety of a module M is the same as that of $\mathrm{Hom}_K(M, K)$, so this also gives the singular variety of $\mathrm{H}_*(X)$.

Recall that the product $\mathcal{A}_1 \times \mathcal{A}_2$ of arrangements $\mathcal{A}_i$ in $\mathbb{C}^{\ell_i}$ is the arrangement $\mathcal{A}$ in $\mathbb{C}^{\ell_1+\ell_2}$ consisting of the hyperplanes $H \times \mathbb{C}^{\ell_2}$ for $H \in \mathcal{A}_1$ and the hyperplanes

$\mathbb{C}^{\ell_1} \times H$ for $H \in \mathcal{A}_2$. Any arrangement can be expressed uniquely as the product of irreducible arrangements. The following well-known remark shows that to compute the singular variety of the Orlik-Solomon algebra as a module over E, it suffices to treat the irreducible case:

Proposition 2.1 *The Orlik-Solomon algebra of a product $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2$ of two arrangements is given by $A(\mathcal{A}) = A(\mathcal{A}_1) \otimes_K A(\mathcal{A}_2)$, the tensor product in the category of graded skew-commutative K-algebras. Thus* $\mathrm{sing}(A(\mathcal{A})) = \mathrm{sing}(A(\mathcal{A}_1)) \times \mathrm{sing}(A(\mathcal{A}_2))$.

Proof. A minimal dependent set of hyperplanes in $\mathcal{A}$, or a minimal set with empty intersection, comes from a similar set either in $\mathcal{A}_1$ or in $\mathcal{A}_2$, proving the first statement. The second follows because $A(\mathcal{A})_1$ is the direct sum of the corresponding spaces for $\mathcal{A}_1$ and $\mathcal{A}_2$. A linear form $x = (x_1, x_2)$ is singular for $A(\mathcal{A})$ if x_i is singular on $A(\mathcal{A}_i)$ for both $i = 1, 2$. ∎

The main result of this section is:

Theorem 2.2 *Let $\mathcal{A}$ be an irreducible hyperplane arrangement with Orlik-Solomon algebra A and elements $e_i \in V := A_1$ corresponding to the hyperplanes of $\mathcal{A}$.*

a) If $\mathcal{A}$ is noncentral then the singular variety of A is V.

b) If $\mathcal{A}$ is central then the singular variety of A is the hyperplane spanned by the elements $e_i - e_j$.

Proof. If the singular variety of the Orlik-Solomon algebra A of an arrangement $\mathcal{A}$ does not contain an element $e \in V$, then A is a free module over the subring $K[e]/e^2$. It follows that the Poincaré polynomial $1+t$ of $K[e]/e^2$ divides the Poincaré polynomial $\pi(A, t)$ of A.

On the other hand, Crapo [1967] (see also Schechtman-Terao-Varchenko [1995, Sect. 2]) shows that if $\mathcal{B}$ is an irreducible central arrangement with deconing $\mathcal{A}$, then

$$\pi(A(\mathcal{B}), t)/(1 + t)_{|_{t=-1}} \neq 0.$$

It follows that in this case the singular variety of $A(\mathcal{A})$ contains everything of degree 1. In particular, if $\mathcal{A}$ is an irreducible noncentral arrangement, we may apply this remark to $\mathcal{B} = c\mathcal{A}$. Part *a*) now follows from Proposition 1.3.

If now $\mathcal{B}$ is an irreducible central arrangement, then the formula $A(\mathcal{B}) = A(d\mathcal{B})[e]$ from Proposition 1.3 implies that the singular variety of $A(\mathcal{B})$ is equal to the singular variety of $E'/I' \cong A(d\mathcal{B})$; that is, it consists of precisely the elements of V' as required. ∎

From Theorem 2.2 and Proposition 2.1 we get the general case:

Corollary 2.3 *The singular variety of the Orlik-Solomon algebra A of any arrangement $\mathcal{A}$ is a linear space of V given in dual coordinates to the canonical basis*

$\{e_H\}_{H\in\mathcal{A}}$ *by the system of equations*

$$\sum_{H\in\mathcal{A}_j} x_H = 0,$$

for each central factor $\mathcal{A}_j$*. In particular, the codimension of the singular variety of* A *equals the number of central factors in an irreducible decomposition of* $\mathcal{A}$*.* ∎

Example 2.4 A central arrangement $\mathcal{A}$ in $\mathbb{C}^\ell$ is called *generic* if no set of ℓ or fewer hyperplanes of $\mathcal{A}$ is dependent. Analogously, a noncentral arrangement is called generic if every set of $\ell+1$ or fewer hyperplanes meet transversely (in particular, they don't meet if the number of hyperplanes is $\ell+1$). In the generic noncentral case it follows immediately from the definition that the Orlik-Solomon ideal I is the $(\ell+1)^{\text{st}}$ power $\mathfrak{m}^\ell$ of the maximal ideal $\mathfrak{m}$ of E. From Proposition 1.3 it follows from this that in the generic central case the Orlik-Solomon ideal is the ℓth power of the maximal ideal of the subalgebra E' generated by the differences $e_i - e_j$ of the generators of E.

The homology module $\mathrm{H}_*(X)$ is, as for every arrangement, given by

$$\mathrm{H}_*(X) = \mathrm{Hom}_K(E/I, K) \cong \mathrm{Hom}_E(E/I, E(n)) = (0 :_E I)(n),$$

the nth twist of the annihilator of I. If $I = \mathfrak{m}^\ell$, then $(0 :_E I)(n) = \mathfrak{m}^{n-\ell+1}$. An explicit computation of the resolution of this ideal is given in terms of Schur functors in Eisenbud and Schreyer [2000, Corollary 5.3]; in particular the resolution is linear.

3 The module $F(\mathcal{A})$

Let $W = V^* = \mathrm{H}_1(X)$ be the dual vector space to V, and let $S = \mathrm{Sym}(W)$ be the symmetric algebra of W, a polynomial ring over K.

As usually stated, the Bernstein-Gel'fand-Gel'fand correspondence (BGG) is an isomorphism between the derived category of bounded complexes of coherent sheaves on $\mathbb{P}(V^*)$ and the derived category of bounded complexes of finitely generated graded modules over $E = \wedge V$. But if one examines the proof one can extract a functor $\mathbf{R}$ from the category of graded modules over S and the category of linear free complexes over E, and also a functor $\mathbf{L}$ from the category of graded E-modules to the category of linear free complexes over S. These functors are equivalences of categories; see Eisenbud and Schreyer [2000, Proposition 2.1].

Starting with a graded E-module P the corresponding complex $\mathbf{L}(P)$ over S is

$$\cdots \longrightarrow S\otimes P_i \longrightarrow S\otimes P_{i+1} \longrightarrow \cdots$$

with differential $1\otimes p \mapsto \sum x_i \otimes e_i p$, where x_i and e_i are dual bases of W and V. Starting with a graded S-module M the corresponding complex $\mathbf{R}(M)$ over E is

$$\cdots \longrightarrow \mathrm{Hom}_K(E, M_i) \longrightarrow \mathrm{Hom}_K(E, M_{i+1}) \longrightarrow \cdots,$$

with differential defined similarly.

Starting from a hyperplane arrangement $\mathcal{A}$, we consider the injective resolution of A as an E-module. Recall that since E is Gorenstein, injective resolutions over E are simply the duals (with respect to E or to K) of free resolutions. Thus the injective resolution of A is the K-dual of the free resolution of $\mathrm{H}_*(X)$. By Theorem 1.1, this free resolution, and with it the injective resolution of A, is linear.

Thus we may define $F(\mathcal{A})$ to be the graded S-module that is mapped by $\mathbf{R}$ to the injective resolution of A as an E-module. The reason for choosing the injective resolution over the free resolution in the definition of $F(\mathcal{A})$ is to make $F(\mathcal{A})$ finitely generated.

The following result, which is Theorem 3.7 of Eisenbud and Schreyer [2000] allows us to derive some basic properties of $F(\mathcal{A})$:

Theorem 3.1 *If M is a graded S-module and P is a finitely generated graded E-module, then $\mathbf{L}(P)$ is a free resolution of M if and only if $\mathbf{R}(M)$ is an injective resolution of P.* ∎

Corollary 3.2 *$F(\mathcal{A})$ is generated over S in degree ℓ and has linear free resolution equal to $\mathbf{L}(A)$. In particular,*

a) $F(\mathcal{A})$ has projective dimension ℓ and $\mathrm{Ext}^{\ell}_S(F(\mathcal{A}), S) = K$.

b) The support of $F(\mathcal{A})$ is a linear space whose codimension is the number of central arrangements in an irreducible decomposition of $\mathcal{A}$.

c) The Hilbert function of $F(\mathcal{A})$ is

$$\sum_{i=0}^{\infty} \dim_K (F(\mathcal{A})_i) t^i = (-1)^{\ell} \frac{\chi(\mathcal{A}, t)}{(1-t)^n}$$

Proof. By Theorem 1.1 the injective resolution of A over E, which is dual to the free resolution of $H_*(X)$, is linear. By Theorem 3.1,

$$\mathbf{L}(A): \quad 0 \longrightarrow S \otimes_K A_0 \longrightarrow \cdots \longrightarrow S \otimes_K A_\ell \longrightarrow F(\mathcal{A}) \longrightarrow 0$$

is a (linear) free resolution of $F(\mathcal{A})$, proving the first statement and computing the projective dimension.

a): The degree 0 and 1 parts of A coincide with those of E; thus the left-hand terms of the resolution above are the same as those in $\mathbf{L}(E)$, the Koszul complex. This allows us to compute the Ext in part a).

b): Aramova, Avramov and Herzog [2000] show in general that the singular variety of an E-module P is the support of the $S = \mathrm{Ext}^*_E(K, K)$-module $\mathrm{Ext}^*_E(P, K)$, which is the same (since E is Gorenstein) as the support of the module $\mathrm{Ext}^*_E(K, P)$. By Eisenbud and Schreyer [2000, Proposition 2.3] this is the module $F(\mathcal{A})$.

c): Knowing the free resolution of $F(\mathcal{A})$ allows us to compute its Hilbert series, just as in the proof of Theorem 1.1. ∎

Example 3.3 If $\mathcal{A}$ is a generic noncentral arrangement of n hyperplanes in K^ℓ, then A is $E/\mathfrak{m}^{\ell+1}$, so the free resolution of $F(\mathcal{A})$ is a truncation of the Koszul complex, and $F(\mathcal{A})$ is isomorphic to the $(v-\ell)^{\text{th}}$ syzygy module of the trivial S-module K. ∎

We have already seen that if $\mathcal{A}$ is a generic noncentral arrangement then the Orlik-Solomon ideal of $\mathcal{A}$ is a power of the maximal ideal of E, and thus has a linear free resolution. We will show that this property characterizes generic arrangements and their cones. We begin with a general result characterizing deformations of powers of the maximal ideal:

Theorem 3.4 *Let $I \subset E$ be an ideal in the exterior algebra. Both I and $(E/I)^*$ admit linear free resolutions if and only if I reduces to a power of the maximal ideal modulo some (respectively any) maximal E/I regular sequence of linear forms of E.*

Proof. If $f_1, \dots, f_s \in E_1$ is a regular sequence on E/I then I and $(E/I)^* = \mathrm{Hom}_K(E/I, K)$ are also free over $K\langle f_1, \dots, f_s\rangle$. The freeness of E/I over $K\langle f_1, \dots, f_s\rangle$ implies that, the image of I in $E/(f_1, \dots, f_s)$ is isomorphic to $I/(f_1, \dots, f_s)I$, and also that the dual of $E/(I + (f_1, \dots, f_s))$ is $(E/I)^* \otimes_E E/(f_1, \dots, f_s)$. Thus the minimal free resolutions of I and $(E/I)^*$ are linear if and only if the minimal resolutions of $I/(f_1, \dots, f_s)I$ and $(E/(I + (f_1, \dots, f_s)))^*$ are linear, and it follows from Eisenbud and Schreyer [2000, Section 5] that if the image of I in $E/(f_1, \dots, f_s)$ is a power of the maximal ideal, then the minimal free resolutions of I and $(E/I)^*$ are linear.

To prove the converse, the argument given above reduces us to showing, in the case where the singular variety of E/I is V, that if the resolutions of I and $(E/I)^*$ are linear, then I is itself a power of the maximal ideal.

Our hypothesis implies in particular that module $(E/I)^*$ is generated in a single degree. It follows by Nakayama's Lemma and duality that the socle of E/I (the annihilator in E/I of $\mathfrak{m}$) is generated in a single degree, say degree s. Thus $I_j = E_j$ for $j > s$, and it suffices to show that $I_j = 0$ for $j \le s$.

By Theorem 3.1, both $\mathbf{L}(E/I)$ and $\mathbf{L}(I^*)$ are free resolutions; let F be the module whose resolution is $\mathbf{L}(E/I)$. By Aramova, Avramov, and Herzog [2000] its support is the singular variety of E/I, that is, V.

Duality (into K) over the exterior algebra gives an exact sequence $0 \to (E/I)^* \to E^* \to I^* \to 0$. Taking duals commutes with the functor $\mathbf{L}$ (up to shifts), so we get an exact sequence of complexes $0 \to \mathbf{L}(E/I)^* \to \mathbf{L}(E)^* \to \mathbf{L}(I^*) \to 0$, where now the duals denote $\mathrm{Hom}_S(-, S)$. The homology of $\mathbf{L}(E/I)^*$ at $S \otimes_K ((E/I)_s)^*$ is $\mathrm{Ext}^0_S(F, S)$, which is nonzero because F has support V. It follows from the exact sequence that $\mathbf{L}(I^*)$ has nonzero homology at the term $S \otimes (I_{s+1})^*$. Since $\mathbf{L}(I^*)$ is a resolution, this must be the last term of the complex—that is, $I_j = 0$ for $j \le s$, as required. ∎

Example 3.5 The ideals characterized in Theorem 3.4 include powers of the maximal ideal in subalgebras generated by linear forms (this will be the case for cones

over hyperplane arrangements) but also many that are *not* of this form. Here is the simplest concrete example: Let

$$I := (ab + cd, ac, bc) \ \subset \ E := K\langle a, b, c, d\rangle.$$

It is easy to check that the three given quadrics form a Gröbner basis with respect to any order with $ab > cd$. Since d is a regular element on E modulo the initial ideal (ab, ac, bc), it follows that d is regular on E/I. It is evident that I reduces modulo d to the square of the maximal ideal. To see that I is not the square of the maximal ideal of any exterior subalgebra on 3 variables, note that the quadrics in 3 variables are all of rank 2, where as I contains an element of rank 4 (here the rank is defined via the identification between elements of E_2 and skew-symmetric 4×4 matrices.)

Corollary 3.6 *The Orlik-Solomon ideal of $\mathcal{A}$ admits a linear free resolution over E if and only if $\mathcal{A}$ is obtained by successively coning a generic noncentral arrangement.*

Proof. We have already seen that the property holds for generic noncentral arrangements. If I is the Orlik-Solomon ideal of $\mathcal{A}$, then the Orlik-Solomon ideal of $c\mathcal{A}$ in $E[e_0]$ is $IE[e_0] = I \otimes_E E[e_0]$, which has free resolution obtained from that of I by tensoring with $E[e_0]$; in particular, the linearity is not affected.

Deconing $\mathcal{A}$ as many times as possible, it now suffices to show that if $\mathcal{A}$ is noncentral and I has a linear resolution then $\mathcal{A}$ is generic. Since $\mathcal{A}$ is noncentral it can have no central factors in its irreducible decomposition, and thus the singular variety of $\mathcal{A}$ is the whole of the vector space V of linear forms.

The theorem now follows from a more general result. Recall from Aramova, Avramov, and Herzog [2000] that a sequence of elements $f_1, \ldots, f_s \in E_1$ is called a regular sequence on an E-module M if M is free over $K\langle f_1, \ldots, f_s\rangle$, or equivalently, if the annihilator of f_i in $M/(f_1, \ldots, f_s)M$ is $f_iM/(f_1, \ldots, f_s)M$ for every i. In this case the minimal free resolution of $M/(f_1, \ldots, f_s)M$ over $E/(f_1, \ldots, f_s)$ is obtained by reducing the minimal E-free resolution of M modulo $(f_1, \ldots, f_s)$. The length of any maximal regular sequence on M is equal to the codimension of the singular variety of M in V.

Remark 3.7 Theorem 3.4 is actually equivalent to the Theorem of Horrocks that characterizes the bundle $\Omega^i_{\mathbb{P}(W)}(i)$ as the unique indecomposable sheaf $\mathcal{F}$ such that the only nonzero intermediate cohomology of any twist of $\mathcal{F}$ is $\mathrm{H}^i(\mathcal{F}) = K$. To see this one uses the correspondence between powers of the maximal ideal of E and the twisted exterior powers of the cotangent sheaf $\Omega^i_{\mathbb{P}(W)}(i)$, as well as the relation between resolutions over E and cohomology of sheaves on $\mathbb{P}(W)$, all explained in Eisenbud and Schreyer [2000].

4 How singular is a singular element?

We keep the notation $W = V^*$ and $S = \mathrm{Sym}(W)$ from the last section. Let N be a graded E-module, such as $\mathrm{H}^*(X)$. Denote the homology of $N \xrightarrow{e} N \xrightarrow{e} N$ by $\mathrm{H}(e, N)$; that is, $\mathrm{H}(e, N) = (\mathrm{Ann}_N\, e)/eN$, so that $\mathrm{H}(e, N) \neq 0$ if and only if e is in the singular variety of N. In general, it is not easy to say in which degrees the homology $\mathrm{H}(e, N)$ will occur, but if N has a linear injective resolution and socle in one degree, say degree 0, as is the case of $\mathrm{H}^*(X)(l)$, then the following result shows that $\mathrm{H}(e, N) \neq 0$ if and only if $e: \ N_1 \to N_0$ fails to be surjective, or equivalently if and only if e annihilates a degree 0 element of N^*.

Theorem 4.1 *Let N be a finitely generated E-module with socle in degree 0 having a linear injective resolution, and let F be the S-module such that the resolution of N is $\mathbf{R}(F)$. For any $e \in V$ we have*

a) For each i,

$$\mathrm{H}(e, N)_i = \mathrm{Tor}_i^{\mathcal{O}_{\mathbb{P},e}}(\kappa(e), \mathcal{F}_e),$$

where $\kappa(e)$ denotes the residue class field of $\mathbb{P} = \mathbb{P}(W)$ at e, and $\mathcal{F}_e$ denotes the stalk at e of the sheaf $\mathcal{F}$ on $\mathbb{P}$ corresponding to the module F.

b) $\mathrm{H}(e, N)_i \neq 0$ if and only if i is between 0 and the projective dimension of $\mathcal{F}_e$ as an $\mathcal{O}_{\mathbb{P},e}$-module. In particular, $\mathrm{H}(e, N) \neq 0$ if and only if $\mathrm{H}(e, N)_0 \neq 0$.

c) Suppose $e \in \mathrm{sing}(N) = X$ is a generic point of the component of X in which it lies. If the codimension of $X \subset \mathbb{P}$ at e is c, then $\mathrm{H}(e, N)_i \neq 0$ precisely for $0 \leq i \leq c$.

Problem 4.2 Characterize, in terms of the arrangement, the subset of $\mathrm{sing}(\mathrm{H}^*(X))$ consisting of those e for which the projective dimension of $\mathcal{F}(\mathcal{A})_e$ is greater than the codimension of the largest component of $\mathrm{sing}(\mathrm{H}^*(X))$ on which e lies. See Yuzvinsky [1995], Falk [1997], Cohen-Suciu [1999] and Libgober-Yuzvinsky [2000] for related computations.

Proof. Everything follows from the formula given in a). To prove this formula, recall that if $t \in W$ is any element outside $e^\perp$, then $\mathcal{F}_e$ may be represented as the degree 0 part of the localization, at $(e^\perp)$, of the module $F[t^{-1}]$. Thus if we write $R = S[t^{-1}]_0$, then the Tor we are concerned with may be represented as $\mathrm{Tor}_i^R(R/(e^\perp)_0, F[t^{-1}]_0)$. Since localizing and taking the degree 0 part are both exact functors, we may rewrite this as $\mathrm{Tor}_i^S(S/(e^\perp)[t^{-1}], F[t^{-1}])_0$. By Theorem 3.1, $\mathbf{L}(N)$ is a free resolution of F, and thus $\mathrm{Tor}_i^S(S/(e^\perp)[t^{-1}], F[t^{-1}])_0$ is the ith homology of $\mathbf{L}(N) \otimes_S R_e$, where $R_e = K[t, t^{-1}]$ has S-module structure coming from the map $S \to K[t, t^{-1}]$ derived from $e : W \to K = Kt$. But $\mathbf{L}(N) \otimes_S R_e$ is nothing but the complex

$$\cdots \xrightarrow{t\otimes e} R_e \otimes_K N_{i-1} \xrightarrow{t\otimes e} R_e \otimes_K N_i \xrightarrow{t\otimes e} R_e \otimes_K N_{i+1} \xrightarrow{t\otimes e} \cdots$$

whose degree 0 part is

$$\cdots \xrightarrow{t\otimes e} Kt^{-i-1} \otimes_K N_{i-1} \xrightarrow{t\otimes e} Kt^{-i} \otimes_K N_i \xrightarrow{t\otimes e} Kt^{-i+1} \otimes_K N_{i+1} \xrightarrow{t\otimes e} \cdots,$$

a complex computing $\mathrm{H}(e, N)_i$. ∎

Example 4.3 Each part of Theorem 4.1 fails for a module whose resolution is linear for d steps, no matter how large d is. To see this, note first that if we have an exact sequence

$$0 \to N' \to F \to N \to 0,$$

such that F is free, then the homology under multiplying by e satisfies $\mathrm{H}(e, N') = \mathrm{H}(e, N)(1)$, Thus a module whose linear syzygy chain ends at a certain point cannot have a minimal generator annihilated by a linear form! Perhaps the simplest example is the dual of the dth module of syzygies of $E/(ab)$, where a, b are independent linear forms.

Restating part of this in the case of a hyperplane arrangement, we get:

Corollary 4.4 *An element $e \in V$ is singular for A (or equivalently for $\mathrm{H}_*(X)$) if and only if there is a nonzero element of $\mathrm{H}_\ell(X)$ annihilated by e.*

Proof. We have shown that $\mathrm{H}^*(X)$ has socle in degree ℓ (note that the signs are opposite to those of Eisenbud and Schreyer [2000]) and linear injective resolution, so $N = \mathrm{H}^*(X)(\ell)$ satisfies the hypothesis of Theorem 4.1. Thus e is singular for N if and only if $e : H^{\ell-1}(X) \to H^\ell(X)$ fails to be surjective, or, dually, $e : H_\ell(X) \to H_{\ell-1}(X)$ fails to be injective. ∎

5 Syzygies of Monomial Ideals in the Exterior Algebra

In this section we give a conceptual description and proof of the correspondence between free resolutions of certain modules over exterior and symmetric algebras first proved by Aramova, Avramov, and Herzog [2000] and Römer [2001]. The main idea is an isomorphism between certain subcategories of the categories of modules over these two algebras. Our approach provides a simple explanation for the shape of the formula relating the corresponding multigraded Betti numbers (Corollary 5.7).

Let V be an n-dimensional vector space over the field K, with basis $x_1, \ldots, x_n$. We will denote by $S = \mathrm{Sym}(V)$ the symmetric algebra over V, which we identify with the ring of polynomials over K in the n variables $x_1, \ldots, x_n$, and by $E = \Lambda(V)$ the exterior algebra of the vector space V. Both these algebras have a natural $\mathbb{Z}^n$ grading in which each monomial (product of the x_i) generates a homogeneous component. (Note that in earlier sections we wrote $S = \mathrm{Sym}(W)$, where W was the dual of V. Since we have explicitly chosen a basis of V we may identify V with W.)

We say that a $\mathbb{Z}^n$-graded module M over E or S is *square-free* if it admits a free presentation $F \longrightarrow G \longrightarrow M \longrightarrow 0$ where each generator of F and G has the degree of a square-free monomial. Note that the presentation map $F \longrightarrow G$ is represented by a matrix whose entries are scalars times monomials. Examples include the Stanley-Reisner rings S/I where I is an ideal generated by square-free monomials, but also such things as the cokernel of the matrix

$$\begin{pmatrix} x_0 & 0 \\ -x_1 & x_1 \\ 0 & -x_2 \end{pmatrix},$$

the canonical module of the cone over 3 points in the plane.

There is a 1-1 correspondence between square-free modules over S and over E obtained by interpreting the presentations as matrices over S or over E; we will write ${}_SM$ and ${}_EM$ for the two.

We can describe the correspondence of resolutions in a simple way as follows:

Start from a free resolution of a square-free module ${}_SM$. Replace each free module in the resolution by a module made from the sum of the vector spaces of its multihomogeneous elements of square-free degree. It turns out—this is the main point—that this complex of vector spaces has the structure both of a complex of S-modules and a complex of E-modules. The modules in this complex are not free, but they have simple and functorial free resolutions. The free resolutions of the E-modules in the complex fit together to make a double complex, whose total complex is the minimal free resolution of ${}_EM$. A similar procedure allows us to pass in the opposite direction.

The correspondence described above works, with appropriate definitions, in a more general setting, in which E is replaced by one of the algebras

$$R_q := \frac{K\{\, x_1, \ldots, x_n \,\}}{(\langle x_j x_i - q x_i x_j \mid 1 \le i < j \le n \rangle + \langle (1-q) x_i^2 \mid 1 \le i \le n \rangle)}$$

where $K\{\, x_1, \ldots, x_n \,\}$ denotes the free K-algebra on $x_1, \ldots, x_n$, and $q \neq 0$. We leave the details of this generalization to the interested reader.

All modules and free resolutions considered will be assumed $\mathbb{Z}^n$-graded. We identify $\mathbb{N}^n \subset \mathbb{Z}^n$ with the set of monomials of S. By the support of a monomial in either E or S, we will mean the collection of variables present in it. A square-free monomial (or multidegree) is an element $\mathbf{a} \in \{0,1\}^n \subset \mathbb{N}^n$, so $\operatorname{supp}(\mathbf{a}) = \{\, x_j \mid a_j \neq 0 \,\}$.

Modules With Square-free Presentation. The following result is due Bruns and Herzog [1995, Theorem 3.1 a)]:

Proposition 5.1 *Let Γ be any set of monomials of S closed under taking least common multiples. If M is an S-module with generators and relations having degrees in Γ, then all the free modules in a minimal free resolution of M have degrees in Γ.*

Proof. We give a new proof using Gröbner bases, which will easily extend to give Proposition 5.3 as well. Let $F \xrightarrow{\phi} G \longrightarrow M \longrightarrow 0$ be a $\mathbb{Z}^n$-graded free presentation with degrees of F and G in Γ. We may replace $F \xrightarrow{\phi} G$ by a map $F' \xrightarrow{\phi'} G$ so that the generators of F' map to a Gröbner basis of $\ker(G \longrightarrow M)$ by using the Buchberger algorithm; this involves adding free generators whose degrees are the least common multiples of pairs of generators already present, and thus still in Γ. Schreyer's theorem (Eisenbud [1995, Theorem 15.10]) shows that in the symmetric case the kernel of ϕ' is generated by elements of degrees equal to the least common multiples of pairs of degrees of generators of F'.

It follows as in Eisenbud [1995, Theorem 20.2], that the minimal presentation of M has also degrees in Γ, and iterating this process we see that the same is true for the whole syzygy chain. ∎

If M is a square-free module in the sense above, then we say that the *square-free part* of M is the module obtained by factoring out all the homogeneous elements of M with non square-free degrees. Thus for example the square-free part of S itself is the factor ring $R := S/(x_1^2, \ldots, x_n^2)$. More generally, if $\mathbf{a}$ is any square-free monomial, then $S(-\mathbf{a})$ has square-free part $R/\operatorname{supp}(\mathbf{a})(-\mathbf{a})$.

Corollary 5.2 *If M is a square-free module over S then the square-free part of M admits a resolution by direct sums of modules of the form $R/\operatorname{supp}(\mathbf{a})(-\mathbf{a})$.* ∎

An analogous result also holds over E:

Proposition 5.3 *If M is a module over E whose generators and relations have square-free degrees, then the square-free part of M admits a finite resolution by modules of the form $E_{\mathbf{a}} := E/\operatorname{supp}(\mathbf{a})(-\mathbf{a})$.*

Proof. Because the generators and relations of M have square-free degrees, we may write M as the cokernel of a map (always $\mathbb{Z}^n$-homogeneous) between finite direct sums of modules of the form $E_{\mathbf{a}}$, and it thus suffices to show that the kernel of such a map is generated in square-free degrees. Using Gröbner bases we may reduce as above to the monomial case. Exactly as in Eisenbud [1995, Lemma 15.1], one shows that all the relations among monomials are generated by those determined by the fact that any monomial $\mathbf{a}$ is annihilated by the variables in the support of $\mathbf{a}$, and the two-at-a time relations coming from the least common multiples ("divided Koszul relations"). The desired result follows. ∎

The Common Subcategory. The category of modules over E and the category of modules over $R = S/(x_1^2, \ldots, x_n^2)$ have much in common. We make one such connection precise as follows:

Let $\mathbf{a}$ and $\mathbf{b}$ be two monomials in E such that $\operatorname{supp}(\mathbf{a}) \subseteq \operatorname{supp}(\mathbf{b})$, and let $E_{\mathbf{a}}$ and $E_{\mathbf{b}}$ be the cyclic E-submodules generated by these monomials. The natural inclusion $E_{\mathbf{b}} \subseteq E_{\mathbf{a}} \subseteq E$ induces a functorial commutative diagram

$$\begin{tikzcd}
E/\operatorname{supp}(\mathbf{a})(-\mathbf{a}) \arrow[r, "\cong"] & E_{\mathbf{a}} \arrow[r, hook] & E \\
E/\operatorname{supp}(\mathbf{b})(-\mathbf{b}) \arrow[u, "\cdot\mathbf{b}\mathbf{a}^{-1}"] \arrow[r, "\cong"] & E_{\mathbf{b}} \arrow[u, hook] \arrow[r, hook] & E \arrow[u, equal]
\end{tikzcd}$$

where the horizontal isomorphisms are defined by sending 1 to the distinguished generator, and the upper left monomorphism is induced by right multiplication in E with $\mathbf{b}\mathbf{a}^{-1}$, the signed exterior monomial such that $(\mathbf{b}\mathbf{a}^{-1})\mathbf{a} = \mathbf{b}$.

The same commutative diagram holds if we replace E by R, and in fact identifying square-free monomials in E with the corresponding monomials in R defines an equivalence of categories. More precisely:

Proposition 5.4 *Let $\mathcal{A}$ denote the K-additive extension of the category of $\mathbb{Z}^n$-graded submodules of R, with morphisms given by inclusions, let $\mathcal{B}$ denote the K-additive extension of the category of $\mathbb{Z}^n$-graded submodules of E, also with morphisms given by inclusions, and let $k\mathbf{Vect}$ be the category of K-vector spaces.*
The above identification of square-free monomials in E with those of R induces an equivalence $\Psi : \mathcal{A} \longrightarrow \mathcal{B}$ of categories whose restriction (via the natural forgetful functors) to the underlying K-vector spaces is the identity functor. In particular, the functor Ψ preserves acyclic complexes. ∎

Notice that if $\mathbf{a}$ is a square-free monomial, then the square free parts of $S(-\mathbf{a})$ is $R/\operatorname{supp}(\mathbf{a})(-\mathbf{a}) = (\mathbf{a})R$, an object of $\mathcal{A}$. Similarly, the square-free part $E_{\mathbf{a}} = E/\operatorname{supp}(\mathbf{a})(-\mathbf{a}) = (\mathbf{a})E$ of $E(-\mathbf{a})$ is an object of $\mathcal{B}$.

Resolutions over S and E. We let $\mathcal{A}_0$ and $\mathcal{B}_0$ be the additive subcategories generated by these modules and the inclusion morphisms $(\mathbf{a})R \subset (\mathbf{b})R$ and $(\mathbf{a})E \subset (\mathbf{b})E$ when $\mathbf{a}|\mathbf{b}$ as monomials in S.

Certain free complexes over S and E correspond to complexes in the categories $\mathcal{A}_0 \cong \mathcal{B}_0$. We describe the connection with S first:

To a given $\mathbb{Z}^n$-graded complex $\mathbf{F}_\bullet$ of free S-modules

$$\mathbf{F}_\bullet : \quad 0 \longrightarrow F_r \longrightarrow \ldots \longrightarrow F_1 \longrightarrow F_0,$$

with generators in square-free degrees, we associate a complex $\operatorname{sf}(\mathbf{F}_\bullet)$ of R-modules, that we may regard as a complex in $\mathcal{A}_0$. Namely we define $\operatorname{sf}(\mathbf{F}_\bullet)$ as the complex of square-free degrees of $\mathbf{F}_\bullet$, that is

$$\operatorname{sf}(\mathbf{F}_\bullet)_i = \oplus_{\mathbf{b}\in\{0,1\}^n} (\mathbf{F}_i)_{\mathbf{b}},$$

for all i, and where the differentials are induced by the differentials of the original complex $\mathbf{F}_\bullet$. It is easy to see that sf defines a functor from the category of $\mathbb{Z}^n$-graded complexes of free S-modules to the category of complexes in $\mathcal{A}_0$.

It follows from Proposition 5.1 that $\mathbf{F}_\bullet$ is square-free acyclic (that is it has no homology in square-free multidegrees) if and only if the complex $\operatorname{sf}(\mathbf{F}_\bullet)$ is acyclic. It is also clear that $\mathbf{F}_\bullet$ is minimal if and only if $\operatorname{sf}(\mathbf{F}_\bullet)$ is minimal.

We have proven:

Proposition 5.5 *The functor* sf *is an equivalence between the category of square-free complexes of free S-modules and the category of complexes in $\mathcal{A}_0$. It preserves minimality and acyclicity.*

Now we turn to complexes over E. For functorial constructions, we will use the divided power algebra. If U is a finitely dimensional graded vector space, we write $D_l(U)$ for the lth-divided power of U. It is convenient to define $D_l(U)$ as the dual of the lth-symmetric power of the dual space, that is $D_l(U) = (\operatorname{Sym}_l(U^*))^*$. The divided powers $D_l(U)$ have "diagonal" maps $D_{l+1}(U) \longrightarrow D_l(U) \otimes U$ which are the

monomorphisms dual to the surjective natural multiplication map in the symmetric algebra $\mathrm{Sym}_l(U^*) \otimes U^* \longrightarrow \mathrm{Sym}_{l+1}(U^*)$.

We can now go from complexes in the category $\mathcal{B}_0$ to free complexes over E using the Cartan Resolution.

Proposition 5.6 *There exists a functor Φ from the category of complexes in $\mathcal{B}_0$ to the category of complexes of free modules over E whose inverse is obtained by taking square-free parts. Φ preserves acyclicity and minimality. Applied to an acyclic complex in $\mathcal{B}_0$ with homology M, the functor Φ provides an E-free resolution of M.*

Proof. We first define Φ on modules in $\mathcal{B}_0$. It associates to a cyclic module $E_{\mathbf{a}} \cong E/\operatorname{supp}(\mathbf{a})(-\mathbf{a})$ the (resolution) $\Phi(E_{\mathbf{a}}) := D(L_{\mathbf{a}}) \otimes E$, where $L_{\mathbf{a}} := \oplus_{x_i \in \operatorname{supp}(\mathbf{a})} k(-\mathbf{e}_i)$ is the $\mathbb{Z}^n$-graded subspace of V spanned by $\operatorname{supp}(\mathbf{a})$, and whose differentials are induced by the diagonals followed by multiplication in E.

More precisely $\Phi(E_{\mathbf{a}})$ is the complex

$$\Phi(E_{\mathbf{a}}): \qquad \ldots \longrightarrow D_2(L_{\mathbf{a}}) \otimes E(-\mathbf{a}) \longrightarrow L_{\mathbf{a}} \otimes E(-\mathbf{a}) \longrightarrow E(-\mathbf{a}),$$

which is a minimal free resolution of the cyclic module $E_{\mathbf{a}}$. We see at once that $E_{\mathbf{a}}$ is the square-free part of $\Phi(E_{\mathbf{a}})$.

If $\mathbf{a}$ and $\mathbf{b}$ are two monomials in E such that $\operatorname{supp}(\mathbf{a}) \subseteq \operatorname{supp}(\mathbf{b})$, then

$$D(L_{\mathbf{b}}) \otimes E(-\mathbf{b}) \xrightarrow{\pi \otimes (\cdot \mathbf{b}\mathbf{a}^{-1})} D(L_{\mathbf{a}}) \otimes E(-\mathbf{a}),$$

where π is the map induced to divided powers by the canonical projection $\pi : L_{\mathbf{b}} \longrightarrow L_{\mathbf{a}}$, is a morphism of chain complexes lifting the inclusion $E_{\mathbf{b}} \subseteq E_{\mathbf{a}} \subseteq E$.

Given a complex $\mathbf{F}_\bullet$ in $\mathcal{B}$, we may apply Φ to obtain a double complex of free E-modules, and we set $\Phi(\mathbf{F}_\bullet)$ to be the total complex of this double complex. Because of the way Φ is defined on each object of $\mathcal{B}_0$, this functor preserves minimality. The spectral sequences of the double complex shows that it also preserves acyclicity. ∎

As Aramova, Avramov, Herzog [2000] and Römer [2001] observe, the existence of such a construction shows that if an S-module M has a linear free resolution over S if and only if the corresponding E module has a linear free resolution over E. Our version of the construction also "explains" these authors' formula for Betti numbers:

Corollary 5.7 *The following equality holds among Poincaré series:*

$$\sum_{i=0}^{\infty} \sum_{\mathbf{a} \in \mathbb{N}^n} \beta^E_{i,\mathbf{a}}({}_E M) t^i \mathbf{u}^{\mathbf{a}} = \sum_{i=0}^{\infty} \sum_{\mathbf{a} \in \mathbb{N}^n} \beta^S_{i,\mathbf{a}}({}_S M) \frac{t^i \mathbf{u}^{\mathbf{a}}}{\prod_{j \in \operatorname{supp}(\mathbf{a})} (1 - t u_j)}$$

where $\beta^E_{i,\mathbf{a}}({}_E M)$ denotes the dimension of the degree $\mathbf{a}$ part of $\mathrm{Tor}^E_i(M, K)$, and similarly for $\beta^S_{i,\mathbf{a}}({}_S M)$.

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